Generating Functions 2

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October 2026

Problem 1

We start by taking the log of $P(z)$ and differentiating:

$$\begin{aligned}
P(z) &= \prod_{n=1}^{\infty} \frac{1}{1-z^n}, \\
\log P(z) &= -\sum_{n=1}^{\infty} \log(1-z^n), \\
\frac{d}{dz}\log P(z) &= \frac{P'(z)}{P(z)} \\
&= -\sum_{n=1}^{\infty} -\frac{nz^{n-1}}{1-z^n}, \\
\frac{zP'(z)}{P(z)} &= \sum_{n=1}^{\infty} \frac{nz^n}{1-z^n} \\
&= \sum_{n=1}^{\infty} nz^n(1+z^n+z^{2n}+\cdots) \\
&= \sum_{n=1}^{\infty}\sum_{k=1}^{\infty} nz^{nk} \\
&= \sum_{n=1}^{\infty} \left(\sum_{j|n} j\right) z^n \\
&= \sum_{n=1}^{\infty} \sigma(n) z^n. \\
zP'(z) &= P(z)\sum_{j=1}^{\infty} \sigma(j) z^j \\
&= \left(\sum_{m=0}^{\infty} p_m z^m\right)\left(\sum_{j=1}^{\infty} \sigma(j) z^j\right).
\end{aligned}$$

The coefficient of z^n on the left is np_n, and on the right is

$$\sum_{j=1}^{n} \sigma(j) p_{n-j},$$

as desired.

Problem 2

We can think of $c_n^{\{1,2\}}$ as counting the number of sequences of 1s and 2s with total sum n. To make a sequence of size n, we can add 1 to a sequence of size $n-1$ or 2 to a sequence of size $n-2$, so the number of compositions satisfies the Fibonacci recurrence. Also, $c_1 = 1$ and $c_2 = 2$, so the initial conditions are the same as F_2 and F_3, so $c_n^{\{1,2\}} = F_{n+1}$.

Problem 4

The parts that are allowed are integers between 1 and r, which is given by

$$z + z^2 + \cdots + z^r = \frac{z(1-z^r)}{1-z}.$$

Since we need k parts, the generating function is

$$C^{(k,r)}(z) = \left(\frac{z(1-z^r)}{1-z}\right)^k = z^k (1-z^r)^k (1-z)^{-k}.$$

To find the closed formula, we can use the binomial theorem:

$$(1-z^r)^k = \sum_{j=0}^{k} (-1)^j \binom{k}{j} z^{rj},$$

$$(1-z)^{-k} = \sum_{m \geq 0} \binom{k+m-1}{k-1} z^m,$$

$$C^{(k,r)}(z) = z^k \left(\sum_{j=0}^{k} (-1)^j \binom{k}{j} z^{rj}\right) \left(\sum_{m \geq 0} \binom{k+m-1}{k-1} z^m\right).$$

A term contributes to z^n when

$$k + rj + m = n,$$

$$m = n - k - rj,$$

so the corresponding binomial coefficient is

$$\binom{k+m-1}{k-1} = \binom{n-rj-1}{k-1}.$$

So

$$c_n^{(k,r)} = \sum_{j=0}^{\lfloor \frac{n-k}{r} \rfloor} (-1)^j \binom{k}{j} \binom{n-rj-1}{k-1}.$$

Problem 7

By Euler's pentagonal theorem,

$$P(z) = \prod_{n\geq 1} \frac{1}{1-z^n} = \left(\prod_{n\geq 1} (1-z^n) \right)^{-1} = \left(\sum_{k\in\mathbb{Z}} (-1)^k z^{k(3k+1)/2} \right)^{-1},$$

$$\left(\sum_{n\geq 0} p_n z^n \right) \left(\sum_{k\in\mathbb{Z}} (-1)^k z^{k(3k+1)/2} \right) = 1.$$

For all $n \geq 1$, the coefficient of z^n on the right is 0, and the coefficient on the left is

$$p_n - p_{n-1} - p_{n-2} + p_{n-5} + p_{n-7} - \cdots,$$

so it must be equal to 0. Moving everything except p_n to the right gives the recurrence

$$p_n = p_{n-1} + p_{n-2} - p_{n-5} - p_{n-7} + \cdots.$$

Problem 8

By Euler's pentagonal theorem,

$$\prod_{n=1}^{\infty} (1-z^n) = \sum_{k=-\infty}^{\infty} (-1)^k z^{k(3k+1)/2}.$$

In our case, $z = 1/10$, so

$$\varphi = \sum_{k=-\infty}^{\infty} (-1)^k 10^{-k(3k+1)/2}.$$

The exponents are

$$1, 2, 5, 7, 12, 15, 22, 26, 35, 40, \ldots,$$

and the coefficients' signs come in pairs:

$$\varphi = 1 - 10^{-1} - 10^{-2} + 10^{-5} + 10^{-7} - \cdots.$$

This only allows the digits $0, 1, 8$, and 9, since adding a positive pair places 1s far out in the decimal expansion (where it was previously 0s), and adding a negative pair places 8s and 9s in far out positions.

Problem 9

For a point $(x_1, \ldots, x_d) \in \mathbb{Z}^d$, the squared distance from the origin is $x_1^2 + \cdots + x_d^2$. For one coordinate x, the generating function is

$$\sum_{x \in \mathbb{Z}} z^{x^2},$$

so for all d coordinates it is

$$\left(\sum_{x \in \mathbb{Z}} z^{x^2} \right)^d.$$

To allow all distances up to n, by Chapter 1 Problem 9, we can divide by $1 - z$:

$$\sum_{n \geq 0} a_n z^n = \frac{1}{1-z} \left(\sum_{x \in \mathbb{Z}} z^{x^2} \right)^d.$$